

Week 3

Overview

- Simplified Lorentz Group Equivariant Block (LGEB) from [1]
- Stack layers to turn random noise + time embedding into time-dependent vector field with optimal transport flow matching objective
- Produce 30 particles with Cartesian 4-momenta $(p_x, p_y, p_z, E/c)$

Modified architecture based on LorentzNet

x = Cartesian particle 4-momenta, ϕ . = learnable functions, $\psi(\cdot) = \text{sgn}(\cdot) \log(|\cdot| + 1)$,
 c = hyperparameter, $N = \#$ particles, $\|\cdot\|, \langle \cdot \rangle$ = Minkowski norm and inner product

LGBE ([1])

$$m_{ij}^l = \phi_e(h_i^l, h_j^l, \psi(\|x_i^l - x_j^l\|), \psi(\langle x_i^l, x_j^l \rangle))$$

$$m_{ij}^l = \phi_m(m_{ij}^l) \cdot m_{ij}^l$$

$$x_i^{l+1} = x_i^l + \frac{c}{N} \sum_{j \in [N]} \phi_x(m_{ij}^l) \cdot |x_i - x_j|$$

$$h_i^{l+1} = h_i^l + \phi_h(h_i^l, \sum_{j \in [N]} \phi_w(m_{ij}^l) m_{ij}^l)$$

Simple LE block this week

$$m_{ij}^l = \phi_e(\psi(\|x_i^l - x_j^l\|), \psi(\langle x_i^l, x_j^l \rangle))$$

$$m_{ij}^l = \phi_m(m_{ij}^l) \cdot m_{ij}^l$$

$$x_i^{l+1} = x_i^l + \frac{c}{N} \sum_{j \in [N]} \phi_x(m_{ij}, g) \cdot |x_j^l - x_i^l|$$

g = global feature vector (currently time embedding $\phi_t(t)$)

Current architecture

```
LorentzFMNet(  
  (layers): ModuleList(  
    (0-11): 12 x FMLorentzLayer(  
      (phi_t): Sequential(  
        (0): Linear(in_features=1, out_features=64, bias=True)  
        (1): SiLU()  
        (2): Linear(in_features=64, out_features=64, bias=True)  
      )  
      (phi_e): Sequential(  
        (0): Linear(in_features=2, out_features=64, bias=False)  
        (1): LayerNorm((64,)), eps=1e-05, elementwise_affine=True)  
        (2): ReLU()  
        (3): Linear(in_features=64, out_features=64, bias=True)  
        (4): ReLU()  
      )  
      (phi_x): Sequential(  
        (0): Linear(in_features=128, out_features=64, bias=True)  
        (1): ReLU()  
        (2): Linear(in_features=64, out_features=1, bias=False)  
      )  
      (phi_m): Sequential(  
        (0): Linear(in_features=64, out_features=1, bias=True)  
        (1): Sigmoid()  
      )  
    )  
  )  
)
```

Current hyperparameters

- 12 layers in the model; sample initial noise from Gaussian
- Train with conditional optimal transport objective

$$\mathbb{E}_{t \sim \mathcal{U}[0,1], q(x_1) \sim \mathcal{J}, p(x_0) \sim \mathcal{N}(0,1)} \|v_t(\phi_t(x_0)) - (x_1 - x_0)\|^2$$

- 1,000 epochs over batches of size 256
- Adam optimizer with learning rate 10^{-3}
- Trained on 124076 gluon jets to avoid zero-padding issue
- Generate 50,000 jets with 30 particles each with 16 Euler integration steps

Training status

- Couldn't provision resources on NRP Nautilus in time - had issues creating a custom Dockerfile
- Should be done by end of today - just using the base CUDA + Torch dockerfile with install instructions in the Pod YAML

Required work

1. Feature processing and normalization in Cartesian coordinates
2. Particle cloud size is currently fixed at 30. Try zero-padding approach like in MPGAN [2], FPCD [3]: sample jet size and ignore masked particles in message-passing steps, ignore in output
3. Experimentally confirm LE by applying boosts to inputs

Possible next steps

1. Global feature vector:

- If training on all jet types: include type in embedding ([4])
- Condition on global jet features (e.g., jet mass, p_T , etc.) ([5, 6, 3, 4, 7]). Train NFs to produce base features if necessary for unconditional inference; or try joint training (like in FPCD [3])
- Richer time embeddings (e.g., sinusoidal embeddings) (like in FPCD [3])
- Include global feature vector in message-passing step
- Update global features after each layer (like in EPiC-GAN [8])

2. Proof for equivariant output (vector field) $\rightarrow$ equivariant probability densities (like in EquiFM [9])

3. FM hyperparameters: objectives (eg. variance-preserving, variance-exploding), number of steps, noise scheduler

References I

- [1] Shiqi Gong et al. “An Efficient Lorentz Equivariant Graph Neural Network for Jet Tagging”. In: *Journal of High Energy Physics* 2022.7 (July 2022), p. 30. ISSN: 1029-8479. DOI: 10.1007/JHEP07(2022)030. (Visited on 06/30/2025).
- [2] Raghav Kansal et al. *Particle Cloud Generation with Message Passing Generative Adversarial Networks*. Jan. 2022. DOI: 10.48550/arXiv.2106.11535. arXiv: 2106.11535 [cs]. (Visited on 02/13/2025).
- [3] Vinicius Mikuni, Benjamin Nachman, and Mariel Pettee. “Fast Point Cloud Generation with Diffusion Models in High Energy Physics”. In: *Physical Review D* 108.3 (Aug. 2023), p. 036025. ISSN: 2470-0010, 2470-0029. DOI: 10.1103/PhysRevD.108.036025. arXiv: 2304.01266 [hep-ph]. (Visited on 03/13/2025).

References II

- [4] Matthew Leigh et al. “Faster Diffusion Model with Improved Quality for Particle Cloud Generation”. In: *Physical Review D* 109.1 (Jan. 2024), p. 012010. DOI: 10.1103/PhysRevD.109.012010. (Visited on 06/27/2025).
- [5] Matthew Leigh et al. “PC-JeDi: Diffusion for Particle Cloud Generation in High Energy Physics”. In: *SciPost Physics* 16.1 (Jan. 2024), p. 018. ISSN: 2542-4653. DOI: 10.21468/SciPostPhys.16.1.018. arXiv: 2303.05376 [hep-ph]. (Visited on 06/23/2025).
- [6] Benno Käch et al. *JetFlow: Generating Jets with Conditioned and Mass Constrained Normalising Flows*. Nov. 2022. DOI: 10.48550/arXiv.2211.13630. arXiv: 2211.13630 [hep-ex]. (Visited on 03/13/2025).
- [7] Erik Buhmann et al. *EPiC-ly Fast Particle Cloud Generation with Flow-Matching and Diffusion*. Sept. 2023. DOI: 10.48550/arXiv.2310.00049. arXiv: 2310.00049 [hep-ph]. (Visited on 03/13/2025).

References III

- [8] Erik Buhmann, Gregor Kasieczka, and Jesse Thaler. “EPiC-GAN: Equivariant Point Cloud Generation for Particle Jets”. In: *SciPost Physics* 15.4 (Oct. 2023), p. 130. ISSN: 2542-4653. DOI: 10.21468/SciPostPhys.15.4.130. arXiv: 2301.08128 [hep-ph]. (Visited on 03/13/2025).
- [9] Yuxuan Song et al. *Equivariant Flow Matching with Hybrid Probability Transport*. Dec. 2023. DOI: 10.48550/arXiv.2312.07168. arXiv: 2312.07168 [cs]. (Visited on 06/25/2025).